# 2.5 Multiple-Angle and Product-to-Sum Formulas

Name _____ Period ____ Date _____

Pre-Calculus 2.5 (30-minute plan) · Student sheet

**I can…**

- use double-angle formulas to rewrite, evaluate, and solve.
- use power-reducing and half-angle formulas.
- use product-to-sum and sum-to-product formulas.

## Part 1 · Notes: fill in the blanks

**Double-angle:** sin 2u = _____ tan 2u = _____

cos 2u = _____ = _____ = _____

**Power-reducing:** $\sin^2 u$ = _____ $\cos^2 u$ = _____

**Half-angle:** sin(u/2) = _____ cos(u/2) = _____ tan(u/2) = _____

The ± sign in a half-angle formula depends on the quadrant of _____.

**Product-to-sum:** sin u cos v = _____

## Part 2 · Examples (work them with the class)

### Example 1

Use a double-angle formula to rewrite 6 sin x cos x.

### Example 2

$\cos u = \frac{5}{13}$ with $\frac{3\pi}{2} < u < 2\pi$. Find sin 2u, cos 2u, and tan 2u.

### Example 3

Find all solutions in $[0, 2\pi)$: $2\cos x + \sin 2x = 0$

## Part 3 · Partner practice

### Partner practice

$\sin u = \frac{3}{5}$ with $\frac{\pi}{2} < u < \pi$. Find $\sin 2u$.